

Yizhou Wu

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Education

- Expected May 2028** **Ph.D. in Electrical Engineering (Medical Imaging & AI)**, *Georgia Institute of Technology*
Graduate GPA: 3.9/4.0
Research focus: machine learning and representation learning methods for medical imaging analysis, with experience in large-scale brain MRI datasets and diverse downstream tasks including tumor segmentation, survival prediction, neurological disease diagnosis, and cross-cohort generalization.
- May 2023** **B.S. in Electrical Engineering (2nd Major: Computer Science)**, *University of Wisconsin–Madison*
Cumulative GPA: 3.7/4.0 | EE Major GPA: 3.9/4.0
Dean's List.

Research Projects

BrainDINO: Self-supervised Foundation Model for Brain MRI Aug 2025 – Present

- Developed a large-scale self-supervised representation learning framework for brain MRI using **52k+ 3D volumes** aggregated from heterogeneous public neuroimaging datasets.
- Designed a DINO-style teacher–student training strategy for slice-wise MRI representation learning with strict subject-level separation to study robustness under cross-dataset distribution shift.
- Evaluated representation transfer using frozen-backbone adaptation under **limited-label regimes** (10–100% supervision), across diverse clinical tasks including tumor segmentation, disease classification, brain age regression, survival prediction, and molecular mutation prediction.
- Released a public GitHub repository with organized downstream evaluation code and reproducible task-level entry points.

Outcome: First-author arXiv preprint; journal submission under review; code: GitHub.

MedDINOv3: Domain-adaptive Vision Foundation Model Jun 2025 – Oct 2025

- Studied domain-adaptive self-supervised pretraining for dense visual representation learning under large appearance and distribution shift.
- Adapted the DINOv3 self-distillation framework and refined a ViT-based architecture with multi-scale token aggregation for large-scale medical imaging datasets.
- Evaluated cross-modality and cross-domain transfer between CT and MRI benchmarks using the **CT-3M dataset** (3.8M+ slices across 16 datasets). *Outcome: arXiv preprint.*

Brain Metastases Detection and Segmentation Dec 2024 – May 2025

- Developed a **3D deformable convolutional neural network** for detecting and segmenting small brain metastases in MRI, improving robustness to scale variation and sparse lesion distribution.

Outcome: First-author paper in Medical Physics (2025).

Reconstruction-guided Self-supervised Segmentation (AnatoMask) Nov 2023 – May 2024

- Proposed reconstruction-guided self-masking strategies for **self-supervised segmentation**.

- Improved robustness and generalization under **limited-label supervision** through reconstruction-guided representation learning.
- *Outcome: ECCV 2024 paper.*

Machine Learning Systems Projects

Generative Sequential Recommendation on Amazon Reviews 2023

2026

PySpark data engineering + TIGER reproduction (RQ-VAE Semantic IDs + generative retrieval); single node: 32 cores / 251 GB RAM / RTX 6000 Ada, no cluster.

- Engineered a single-node **PySpark** pipeline over **59.77M** Amazon reviews (deduplicated to 59.14M), cutting the working set **26 GB** $\rightarrow$ **3.6 GB** via explicit schemas and applying iterative 5-core filtering (22M $\rightarrow$ 2.39M users; 3.5M $\rightarrow$ 715k items) to build an 18.04M/2.39M/2.39M leave-one-out benchmark.
- Diagnosed lazy-DAG recomputation in iterative k-core filtering: uncached, round 6 alone took **188 s** (**15 $\times$** a cached round) and round 7 was killed past 250 s; adding `.cache()` converged all **8 rounds in 237 s**.
- Quantified skew rather than assuming it — a **1160 $\times$** hot-item ratio (13,970 vs. median 12) produced only **1.05 $\times$ /1.18 $\times$** row imbalance across 256 partitions — and converted a failed 3.5M-row broadcast (>1 GB, over driver `maxResultSize`) into a **BroadcastHashJoin 1.37 $\times$ faster** than `SortMergeJoin` by pre-filtering to 715k items, verified via `.explain()`; reported AQE as a null result (7.0s vs. 8.2s).
- Trained an **RQ-VAE** (768 $\rightarrow$ 32 encoder; 3 residual codebooks $\times$ 256; straight-through estimator, k-means initialization, dead-code revival) over **715,729** frozen Sentence-T5 embeddings, cutting reconstruction loss **0.67** $\rightarrow$ **0.42** at **100% codebook utilization** across all three levels (no collapse), and resolved 231.6k collisions among 484k unique 3-code combinations with a 4th token.
- Designed the generative retrieval stage on top of the resulting Semantic IDs: a **T5 encoder-decoder** over semantic token sequences with **constrained beam search** restricted to valid in-catalog ID prefixes, evaluated by Recall@K / NDCG@K on the leave-one-out split.

Publications

- [1] **Wu, Y.**, Li, Y., Hu, M., Chang, C.-W., Wang, S., Safari, M., Veeraraghavan, H., Qiu, R. L. J., Tian, Z., & Yang, X. (2026). *BrainDINO: A Brain MRI Foundation Model for Generalizable Clinical Representation Learning*. arXiv preprint, arXiv:2604.27277. *Journal submission under review.*
- [2] **Wu, Y.**, Li, Y., Hu, M., Chang, C.-W., Qiu, R. L. J., Wang, T., Shu, H.-K., Mao, H., Tian, Z., & Yang, X. (2025). *Detection and Segmentation of Brain Metastases on MRI Using 3D-MedDCNet*. *Medical Physics*, 52(7), e18001.
- [3] Li, Y., Luan, T., **Wu, Y.**, Pan, S., Chen, Y., & Yang, X. (2024). *AnatoMask: Enhancing Medical Image Segmentation with Reconstruction-guided Self-masking*. *European Conference on Computer Vision (ECCV)*, 146–163.
- [4] Li, Y., **Wu, Y.**, Lai, Y., Hu, M., & Yang, X. (2025). *MedDINOv3: How to Adapt Vision Foundation Models for Medical Image Segmentation?* arXiv preprint, arXiv:2509.02379.
- [5] Li, Y., Wynne, J. F., **Wu, Y.**, Qiu, R. L. J., Tian, S., Wang, T., Patel, P. R., Yu, D. S., & Yang, X. (2025). *Automatic Medical Imaging Segmentation via Self-supervising Large-scale Convolutional Neural Networks*. *Radiotherapy and Oncology*, 204, 110711.
- [6] Zhao, F., **Wu, Y.**, Hu, M., Chang, C.-W., Liu, R., Qiu, R., & Yang, X. (2025). *Current Progress of Digital Twin Construction Using Medical Imaging*. *Journal of Applied Clinical Medical Physics*, 26(9), e70226.

Professional Service

Reviewer for *IEEE Transactions on Medical Imaging* (2025); *MICCAI* (2025); *Medical Physics* (2026)

Industry Experience

MRI Scout-based Anatomical Detection for Safety-Critical Systems May 2024 – Aug 2024, GE Healthcare
MR Application / PSD Development Engineer Intern

Developed safety-critical 3D computer vision models for robust anatomical detection under partial observability and truncated fields of view; queried internal clinical imaging databases using **SQL** to identify and retrieve MRI cohorts for model development and validation; 3D MRI SAR GradWarp correction for reconstruction; system-level implementation in **C++**; built end-to-end data pipelines including data collection, labeling, training, evaluation, and deployment; containerized deployment using **Docker** with reproducible ML workflows under strict reliability constraints.

Outcome: Deployed as an internal safety modeling component at GE Healthcare.

Technical Skills

Programming	Python, C/C++, MATLAB, SQL
Deep Learning	PyTorch, TensorFlow; Vision Transformers (ViT), convolutional neural networks (CNN); self-supervised learning, contrastive learning, masked modeling, self-distillation, transfer learning; residual vector quantization (RQ-VAE), straight-through estimation, sequential recommendation, generative retrieval
Evaluation & Analysis	representation evaluation (CKA), embedding visualization (t-SNE), robustness analysis, statistical validation
Medical Imaging	multimodal medical imaging analysis, survival modeling (Cox regression), cross-domain transfer
Systems & Data	Linux/Unix, Git, Docker, reproducible ML pipelines; PySpark (partition/shuffle tuning, join strategy, physical-plan inspection), Parquet

Teaching Experience

2022–2023 Undergraduate Teaching Assistant, University of Wisconsin–Madison: ECE 532 Matrix Methods in Machine Learning; ECE 252 Introduction to Computer Engineering.